# 高级工程数学
# (Math4AI / 人工智能的数学基础)

2026-09-27

**沈超敏**

**计算机科学与技术学院**

cmshen@cs.ecnu.edu.cn

**文附楼 105**

# 先入群，实名



该二维码7天内（〈O月〈日前）
有效，重新进入将更新

# 正式讲课前 · · · · · ·

微信群（实名）、选课名单、课代表

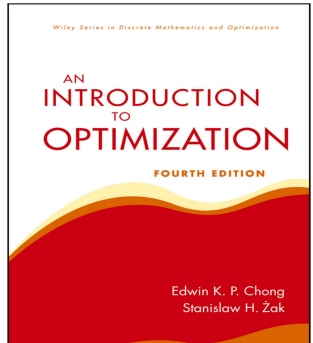
1. 9.27 (Sun)
2. 10.4 (Sun)
3. 10.6 (Tue)
4. 10.11 (Sun)
5. 10.18 (Sun)
6. 10.25 (Sun)
7. 11.1 (Sun)
8. 11.8 (Sun)
9. 11.15 (Sun)

# 教科书

An Introduction to Optimization (Fourth Edition)

作者: Edwin K.P. Chong & Stanislaw H. Zak

出版社: Wiley, 2013 (电子版)



# 今天内容

- **开场白 (第一时间段)**

  一些具体的 logistics: 上课/考试形式、课件、微信群等

  为什么要开这门课？

  本课程与整个研究生学习的关系

  这门课具体内容

  学习方法

- **正式内容: Ch. 1**
- Ch. 2 (视时间)

# 引子：《双城记》

It was the best of times, it was the worst of times,

it was the age of wisdom, it was the age of foolishness···

**这是最好的时代，也是最坏的时代；**

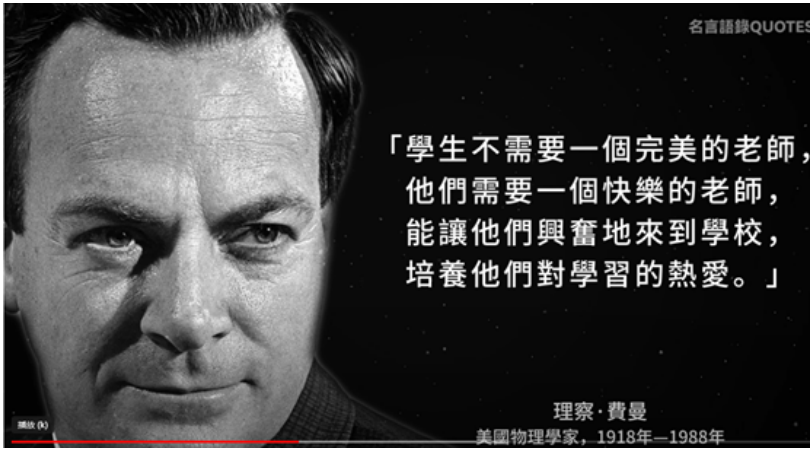
**这是智慧的时代，也是愚昧的时代；**

**这是信仰的时期，也是怀疑的时期；**

**这是光明的季节，也是黑暗的季节...**

Bill Thurston：**数学的本质是推进人类的理解，而不仅仅是制造产出**

# 学习方法




大疑大悟、小疑小悟、不疑不悟　　《金刚经》

# 问学

例：http://faculty.bicmr.pku.edu.cn/~dongbin/Teaching.html

## Principles of Imaging – Unified Perspectives

Bin Dong & GPT-4

2023.09.09

**Table of Content:**


1. Introduction to Imaging Principles and Techniques
   1.1 Role of Imaging in Different Fields
   1.2 Fundamental Concepts in Imaging
2. Basics of Electromagnetic (EM) Radiation and Mechanical Waves
   2.1 Electromagnetic Waves
   2.2 Wave-Particle Duality and Imaging
   2.3 Mechanical Waves
3. Wave Interaction Modalities and Imaging: Transmission, Absorption, Reflection, Scattering, Diffraction
4. Imaging Modalities Based on Transmission
   4.1 X-ray Imaging and Computed Tomography (CT)

# 反向学习法

## 陶哲轩力荐，哈佛反向学习法火了：教会AI就是教会自己

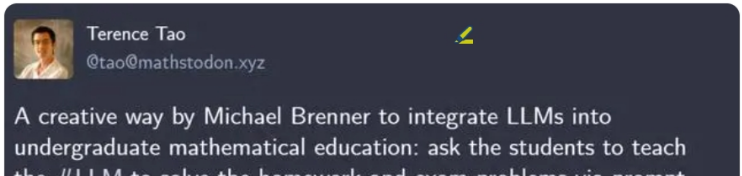
量子位　2024-09-02 12:46 北京


AI 摘要

哈佛教授Brenner提出创新教学法，让学生教AI完成数学作业，以锻炼学生问题拆解和提示词工程技术。他另一门课将烹饪与科学结合，打破学生固有认知。

摘要由作者通过智能技术生成　有用 | 


数学大佬陶哲轩力荐，哈佛反向学习法火了：**教会AI就是教会自己。**



数学分析　Why

高等数学　How

高中毕业时的数学水平：400 年前

学完高数的数学水平：150 年前

学完本课程后的数学水平：50 年前（部分内容前沿）

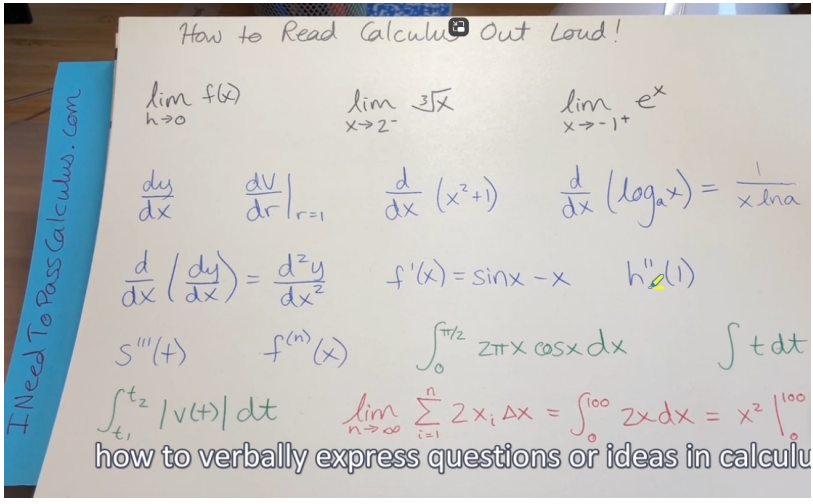



232132132132132132132132132132321321+9988998987766655544332=?



232,132,132,132,132,132,132,132,132,132,132,132,321,321 + 9,988,998,987,766,655,544,332 equals 9,988,998,987,998,787,687,006

# 数学符号如何读？



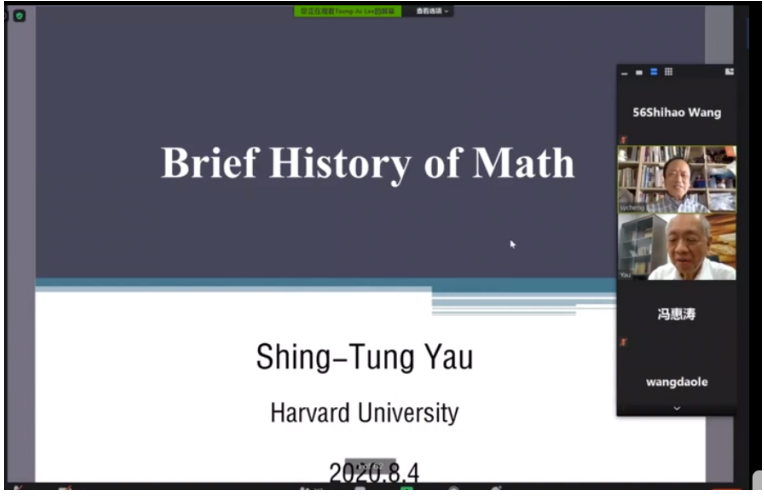

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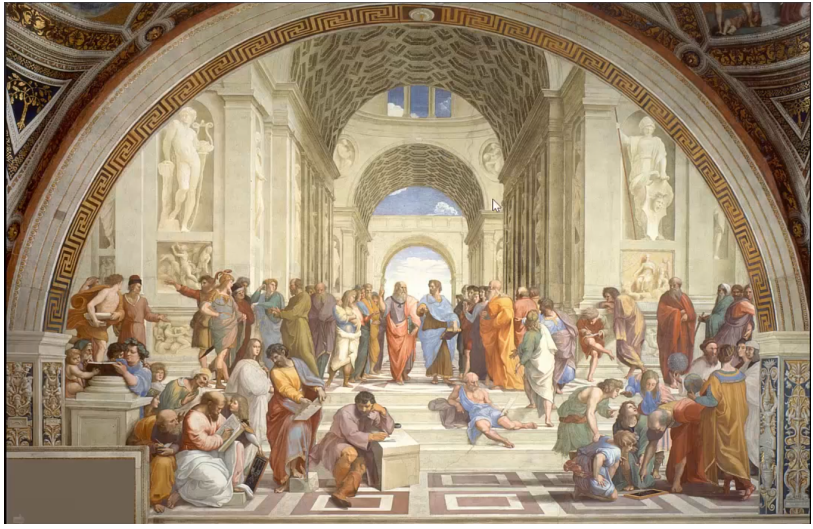
跟随伟大人物的思想，是一门最引人入胜的科学

普希金

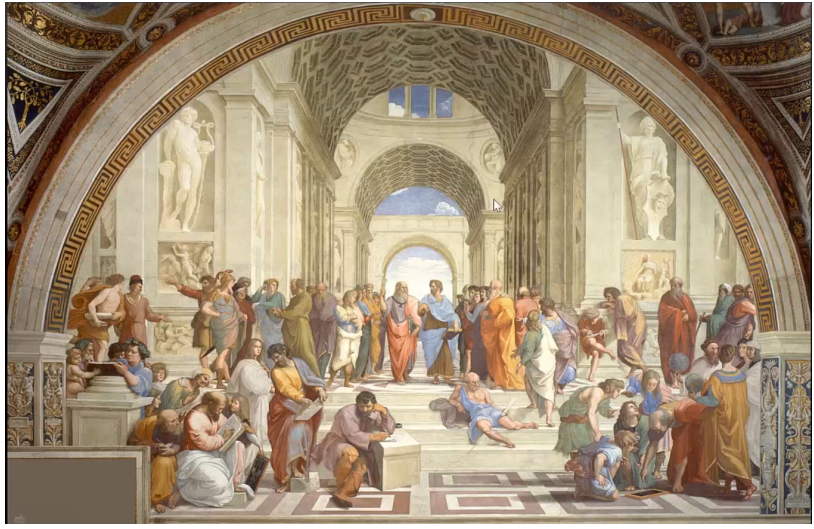
# 数学史

https://www.bilibili.com/video/BV1Ef4y197wQ?from=search&seid=5832620057198290064

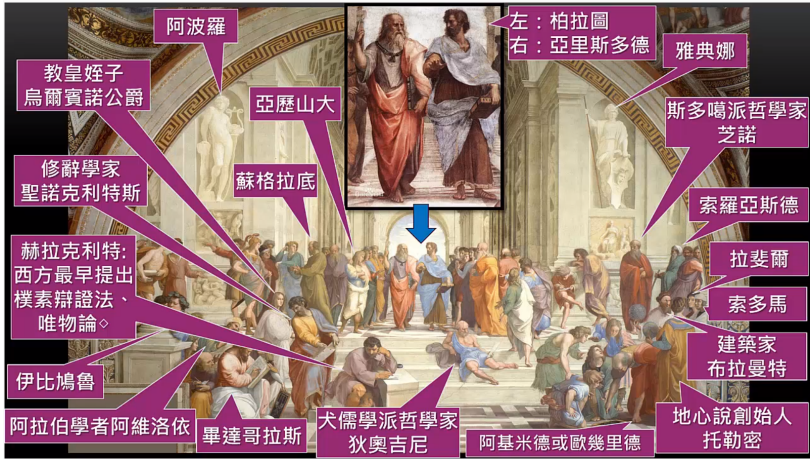




School of Athens (雅典学院)



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School of Athens (雅典学院)

如果不知道远溯古希腊各代前辈
所建立和发展的概念、方法和结果,
我们就不可能理解
近五十年来数学的目标,
也不可能理解它的成就.

—— 外尔 (C. H. H. Weyl, 1885–1955)


ATLANTIC OCEAN
EUROPE
GAUL
ALPS
Rhone R.
PYRENEES MTS.
IBERIA
Marseilles
CORSICA
BALEARIC ISLANDS
SARDINIA
Rome
ITALY
ADRIATIC SEA
TYRRHENIAN SEA
Tarentum
MEDITERRANEAN SEA
SICILY
Syracuse
IONIAN SEA
Danube R.
BLACK SEA
CAUCASUS MTS.
Byzantium
GREECE
Chaeronea
Corinth
Sparta
Athens
Gulf of Corinth
Aegean Sea
SAMOS
Miletus
ASIA MINOR
MELOS
RHODES
CRETE
CYPRUS
Euphrates R.
SYRIA
Damascus
Jerusalem
CYRENAICA
Naucratis
Memphis
EGYPT
Nile R.
RED SEA
AFRICA
500 miles
GREECE AND ITS COLONIES IN 550 B.C.

**數學詞源**: **希腊语** $\mu\alpha\theta\eta\mu\alpha\tau\iota\kappa\zeta$ (Mathematikós),
**意思是“学问的基础”，源于** $\mu\alpha\theta\eta\mu\alpha$ (máthema，“科学，知识，学问”)

Thales (624 B.C. – 546 B.C.)　　**半圆内切的角度是直角**

Pythagoras (570 B.C. – 495 B.C.)　**“数学”一词的专门化使用是源于毕达哥拉斯的想法**

Euclid (325 B.C. – 265 B.C.)　**公理化方法**

Archimedes (287 B.C. – 212 B.C.)

.....

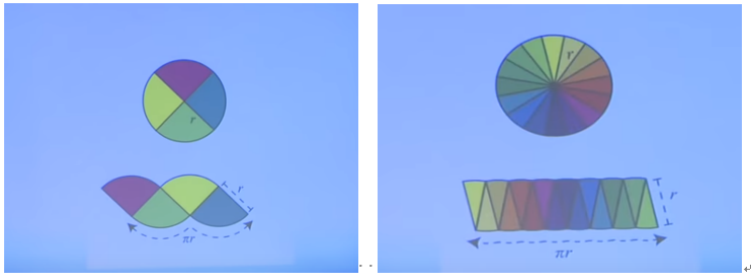
**有史以来最重要的 3 个数学家**: Archimedes、Newton、(Euler/Gauss)

**微积分的创立者**: Newton、Leibnitz (Leibniz / von Leibniz)

# Archimedes 的贡献

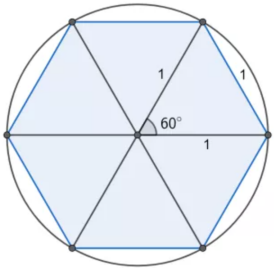
**已知**：$\pi$ 的定义为 $\pi = C/d$，i.e., $C = 2\pi r$,

**求证**：$S = \pi r^2$




https://www.youtube.com/watch?v=1r6893ga_So

# 东京大学入学考试题 (2003)：证明 $\pi > 3.05$




比如，在一个半径为1的圆里面做一个正六边形，可知圆的周长大于正六边形的周长，也就是：

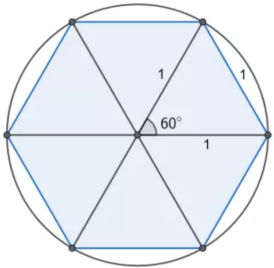
$$2\pi > 6$$

因此：

$$\pi > 3$$

思考：或用其它方法求得更精确的 $\pi$ 值

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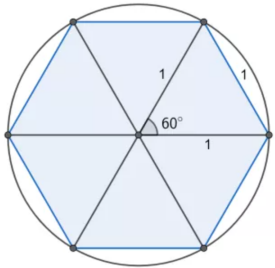
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因此：

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**思考：或用其它方法求得更精确的 $\pi$ 值**

# 数学史

https://www.bilibili.com/video/BV1Ef4y197wQ?from=search&seid=5832620057198290064

已经 85 岁的吴先生思想十分活跃，他兴致勃勃地谈起科技发展的观点和数学的研究方向。

他说：“从‘文化大革命’以后，我的研究方向有一个很大的转变，这是受中国数学的启发。中国传统科学中，也有珍贵的宝藏。要把东西方优秀的成果结合起来，为我所用。学习中国数学的历史，我发现中国数学发展的途径、思想方法和现在的科学理论是相反的，是根本不相同的。中国是算法式的。算法是你做了第一步，就知道第二步该怎么做，做了第二步就知道第三步该怎么做。而西方的现代数学，每一步的证明都要经过思考，走了第一步不知道第二步怎么做。我受了这个启发，就想是不是在数学的许多领域也可以按照中国这种算法的方式。这在现在的实现条件，就是我们有计算机。”

我说：“有人这么概括说，计算机和数学的结合是思维和工具的结合，是科学和机械的结合。”

# 如何证明负负得正？

**可以运用**：**分配律**、-1+1=0

# Ch. 1. Methods of Proof and Some Notations

## 1.1 Method of Proof

## 1.2 Notation

Exercises

# 逻辑

税。有一次平原君家不肯交租，赵奢依法办理这事件，杀了平原君家九个主事的人。平原君发怒，准备杀赵奢。赵奢趁机劝说道：“您在赵国是贵公子，今天纵容您家不奉公守法，那么国法就会削弱，国法削弱那么国家就会衰弱；国家衰弱那么各诸侯国就会进兵来侵犯；各诸侯国进兵来侵犯，赵国就不能存在了，您怎么还能得到这样的富足呢？凭着您这样高贵的地位，能够奉公守法，那么全国上下就会公平；上下公平，国家就强盛；国家强盛，赵国的统治就稳固。而您作

错误类型：Slippery Slope Fallacy

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**两个命题** A, B

(**True** or **False**)

运算符 **and**, **or**, **not**

可组成 A **and** B

A **or** B

**not** A

**not**(A **and** B)

...

| A | B | A and B | A or B |
|---|---|---|---|
| F | F | F | F |
| F | T | F | T |
| T | F | F | T |
| T | T | T | T |

| A | not A |
|---|---|
| F | T |
| T | F |

| A | B | A and B | not(A and B) |
|---|---|---|---|
| F | F | F | T |
| F | T | F | T |
| T | F | F | T |
| T | T | T | F |

| A | B | not A | not B | (not A) or (not B) |
|---|---|---|---|---|
| F | F | T | T | T |
| F | T | T | F | T |
| T | F | F | T | T |
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DeMorgan's Law

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| T | F | F | T |
| T | T | T | F |

| A | B | not A | not B | (not A) or (not B) |
|---|---|---|---|---|
| F | F | T | T | T |
| F | T | T | F | T |
| T | F | F | T | T |
| T | T | F | F | F |

not (A and B) $\iff$ (not A) or (not B)

DeMorgan's Law

# 1.1. Method of Proof

两个命题 A, B

(**True** or **False**)

运算符 **and**, **or**, **not**

可组成 A **and** B

A **or** B

**not** A

**not**(A **and** B)

...

| A | B | A and B | A or B |
|---|---|---|---|
| F | F | F | F |
| F | T | F | T |
| T | F | F | T |
| T | T | T | T |

| A | not A |
|---|---|
| F | T |
| T | F |

| A | B | A and B | not(A and B) |
|---|---|---|---|
| F | F | F | T |
| F | T | F | T |
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| T | T | F | F | F |

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DeMorgan's Law

# 剖析 $A \to B$

| A | B | $A \to B$ |
|---|---|---|
| 真 | 真 | 真 |
| 真 | 假 | 假 |
| 假 | 真 | 真 |
| 假 | 假 | 真 |

# 1.1. Method of Proof (续)

A implies B　　$A \Rightarrow B$

$\Updownarrow$

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上述表达式可以理解为: A 分成 2 种情况. 1) A 成立, 则B 成立;
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$A \Leftrightarrow B$ simply means $(A \Rightarrow B)$ and $(B \Rightarrow A)$

A if and only if B, 或 A is equivalent to B

$A \Rightarrow B \iff (\text{not } B) \Rightarrow (\text{not } A)$ (逆否命题)

$\Rightarrow$

如果在 A 推出 B 的前提下, 而假设 not B 得出 not A 不成立, 则 A 成立, 但 A 成立意味着 B 成立, 同 not B 矛盾

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# 1.1. Method of Proof (续)

如何证明 $A \Rightarrow B$?

1. The direct method
2. Proof by contraposition (证明逆否命题)
3. Proof by contradiction
4. Principle of induction

# 数学归纳法需要满足 2 个条件

**例 (选自华罗庚科普著作选)** 所有的正整数都相等.

这个命题显然是荒谬的.

但是如果我们丢开“当 $n = 1$ 的时候，这个命题是正确的”不管，那末可以用“数学归纳法”来“证明”它.

这里，第 $k$ 号命题是：“第 $k - 1$ 个正整数等于第 $k$ 个正整数”，就是

$$k - 1 = k$$

两边都加上 $1$，就得

$$k = k + 1$$

这就是说，第 $k$ 个正整数等于第 $k + 1$ 个正整数. 这不是说明了所有的正整数都相等了吗？

错误就在于，我们没有考虑 $k = 1$ 的情况

# 1.2. Notation

- 标量用小写 (非黑体) 字母表示，如 $x$;
  向量用小写 (黑体) 字母表示，如 $\boldsymbol{x}$; $\boldsymbol{x} = (x_1, \cdots, x_n)$;
  矩阵用大写字母表示，如 $A$;
- 集合用大写 (花写) 字母表示，如 $X$，$\mathcal{X}$; 集合中的元素用小写字母表示，如 $X = \{x_1, \cdots, x_n\}$; $X = \{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n\}$;

# Ch. 2. Vector Spaces and Matrices

2.1 Vector and Matrix

2.2 Rank of a Matrix

2.3 Linear Equations

2.4 Inner Products and Norms

Exercises

# 2.1. Vector and Matrix

A column $n$-vector: an array of $n$ numbers

$$\boldsymbol{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

$a_i$: the $i$th component of the vector $\boldsymbol{a}$.

A row $n$-vector

$$[a_1, a_2, \cdots, a_n].$$

# 2.1. Vector and Matrix

Transpose: If

$$\boldsymbol{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix},$$

then

$$\boldsymbol{a}^\top = [a_1, \cdots, a_n]$$

向量的相等: 两个向量的对应的元素均相等

# 2.1. Vector and Matrix

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向量的相等: 两个向量的对应的元素均相等

# 2.1. Vector and Matrix

The operation of addition of vectors has the following properties:

1. The operation is commutative:

$$\boldsymbol{a} + \boldsymbol{b} = \boldsymbol{b} + \boldsymbol{a}.$$

2. The operation is associative:

$$(\boldsymbol{a} + \boldsymbol{b}) + \boldsymbol{c} = \boldsymbol{a} + (\boldsymbol{b} + \boldsymbol{c}).$$

3. There is a zero vector $\mathbf{0} = [0, 0, ..., 0]^T$ such that

$$\boldsymbol{a} + \mathbf{0} = \mathbf{0} + \boldsymbol{a} = \boldsymbol{a}.$$

# 2.1. Vector and Matrix

The vector

$$[a_1 - b_1, a_2 - b_2, \cdots, a_n - b_n]^\top$$

is called the difference between $\boldsymbol{a}$ and $\boldsymbol{b}$ and is denoted $\boldsymbol{a} - \boldsymbol{b}$.

The vector $\mathbf{0} - \boldsymbol{b}$ is denoted $-\boldsymbol{b}$.

$$\boldsymbol{b} + (\boldsymbol{a} - \boldsymbol{b}) = \boldsymbol{a}$$

$$-(-\boldsymbol{b}) = \boldsymbol{b}$$

$$-(\boldsymbol{a} - \boldsymbol{b}) = \boldsymbol{b} - \boldsymbol{a}$$

# 2.1. Vector and Matrix

**数乘**: We define an operation of multiplication of a vector $\boldsymbol{a} \in \mathbb{R}^n$ by a real scalar $\alpha \in \mathbb{R}$ as

$$\alpha \boldsymbol{a} = [\alpha a_1, \cdots, \alpha a_n]^\top.$$

1. The operation is distributive: for any real scalars $\alpha$ and $\beta$,

$$\alpha(\boldsymbol{a} + \boldsymbol{b}) = \alpha \boldsymbol{a} + \alpha \boldsymbol{b},$$
$$(\alpha + \beta)\boldsymbol{a} = \alpha \boldsymbol{a} + \beta \boldsymbol{a}.$$

2. The operation is associative:

$$\alpha(\beta \boldsymbol{a}) = (\alpha\beta)\boldsymbol{a}.$$

3. The scalar 1 satisfies $1\boldsymbol{a} = \boldsymbol{a}$.

4. Any scalar $\alpha$ satisfies $\alpha \mathbf{0} = \mathbf{0}$.

5. The scalar 0 satisfies $0\boldsymbol{a} = \mathbf{0}$.

# 杨振宁教授在清华的《普通物理》

https://www.bilibili.com/video/BV1Fx411T7sV?p=3

(from 第 15 分钟)

# 2.1. Vector and Matrix

$$\alpha \boldsymbol{a} = \mathbf{0} \quad \Longleftrightarrow \quad \alpha = 0 \text{ or } \boldsymbol{a} = \mathbf{0}.$$

$\alpha \boldsymbol{a} = \mathbf{0}$ is equivalent to $\alpha a_1 = \alpha a_2 = \cdots = \alpha a_n = 0$.

**Proof**

$\Leftarrow$ If $\alpha = 0$ or $\boldsymbol{a} = \mathbf{0}$, then $\alpha \boldsymbol{a} = \mathbf{0}$. $\surd$

$\Rightarrow$

(1) If $\boldsymbol{a} \neq \mathbf{0}$, then at least one of its components $a_k \neq 0$. For this component, $\alpha a_k = 0$, and hence we must have $\alpha = 0$.

(2) If $\alpha \neq 0$, 类似

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# 思考题

**解释下列现象**

例如二维空间中的 (1,2) 和 (2,1)，在换了一组基 (1,1), (0,1) 的情况下，它们的坐标分别变成了 (1,1) 和 (2,-1)，此时内积从 4 变成了 1。

这个过程哪里出问题了？

# 2.1. Vector and Matrix

## Basis

Given a subspace $\mathcal{V}$, any set of linearly independent vectors $\{\boldsymbol{a}_1, ..., \boldsymbol{a}_k\} \subset \mathcal{V}$ such that $\mathcal{V} = \text{span}[\boldsymbol{a}_1, ..., \boldsymbol{a}_k]$ is referred to as a basis of the subspace $\mathcal{V}$.

## The dimension of $\mathcal{V}$

All bases of a subspace $\mathcal{V}$ contain the same number of vectors. This number is called the dimension of $\mathcal{V}$, denoted $\dim \mathcal{V}$.

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# 2.1. Vector and Matrix

**Proposition 2.2** If $\{\boldsymbol{a}_1, ..., \boldsymbol{a}_k\}$ is a basis of $\mathcal{V}$, then any vector $\boldsymbol{a}$ of $\mathcal{V}$ can be represented uniquely as

$$\boldsymbol{a} = \alpha_1 \boldsymbol{a}_1 + \cdots + \alpha_k \boldsymbol{a}_k,$$

where $\alpha_i \in \mathbb{R}$, $i = 1, 2, ..., k$.

**Proof**. To prove the uniqueness of the representation of $\boldsymbol{a}$ in terms of the basis vectors, assume that

$$\boldsymbol{a} = \alpha_1 \boldsymbol{a}_1 + \cdots + \alpha_k \boldsymbol{a}_k,$$

and

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Then

$$(\alpha_1 - \beta_1)\boldsymbol{a}_1 + \cdots + (\alpha_k - \beta_k)\boldsymbol{a}_k = 0.$$

Because $\{\boldsymbol{a}_1, \cdots, \boldsymbol{a}_k\}$ is linearly independent, $\alpha_i = \beta_i$. □

# 2.1. Vector and Matrix

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# 2.1. Vector and Matrix

Suppose that we are given a basis $\{\boldsymbol{a}_1, ..., \boldsymbol{a}_k\}$ of $\mathcal{V}$ and a vector $\boldsymbol{a} \in \mathcal{V}$ such that

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The coefficients $\alpha_i$, $i = 1, ..., k$, are called the coordinates of $\boldsymbol{a}$ with respect to the basis $\{\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_k\}$.

The natural basis for $\mathbb{R}^n$ is the set of vectors

$$\boldsymbol{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \boldsymbol{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \cdots, \boldsymbol{e}_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

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Matrix

A matrix is a rectangular array of numbers, commonly denoted by uppercase bold letters (e.g., $A$). A matrix with $m$ rows and $n$ columns is called an $m \times n$ matrix, and we write

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Transpose

$$A^\top = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}$$

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$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Let us denote the $k$-th column of $A$ by $\boldsymbol{a}_k$, then $A = [\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n]$.

The maximal number of linearly independent columns of $A$ is called the rank of the matrix $A$, denoted $\operatorname{rank} A$. Note that $\operatorname{rank} A$ is the dimension of $\operatorname{span}[\boldsymbol{a}_1, ..., \boldsymbol{a}_n]$.

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**Proposition 2.3.** The rank of a matrix $A$ is invariant under the following operations:

1. Multiplication of the columns of $A$ by nonzero scalars.

2. Interchange of the columns.

3. Addition to a given column a linear combination of other columns.

**Proof.**

1. Let $\boldsymbol{b}_k = \alpha_k \boldsymbol{a}_k$, where $\alpha_k \neq 0$, $k = 1, \cdots, n$, and let $B = [\boldsymbol{b}_1, \cdots, \boldsymbol{b}_n]$. Obviously,

$$\operatorname{span}[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n] = \operatorname{span}[\boldsymbol{b}_1, \cdots, \boldsymbol{b}_n],$$

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So, $\alpha_1\boldsymbol{b}_1 + \alpha_2\boldsymbol{b}_2 + \cdots + \alpha_n\boldsymbol{b}_n = \alpha_1\boldsymbol{a}_1 + (\alpha_2 + \alpha_1 c_2)\boldsymbol{a}_2 + \cdots + (\alpha_n + \alpha_1 c_n)\boldsymbol{a}_n$,.
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On the other hand

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# 2.2. Rank of a Matrix

**对方阵** $A$, **可定义** determinant, **记为** $\det A$ or $|A|$.

Determinant **具有如下性质**

1. The determinant of the matrix $A = [\boldsymbol{a}_1, ..., \boldsymbol{a}_n]$ is a linear function of each column; that is,

$$\begin{aligned}&\det[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_{k-1}, \alpha\boldsymbol{a}_k^{(1)} + \beta\boldsymbol{a}_k^{(2)}, \boldsymbol{a}_{k+1}, \cdots, \boldsymbol{a}_n] \\ &= \alpha \det[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_k^{(1)}, \boldsymbol{a}_{k+1}, \cdots, \boldsymbol{a}_n] + \beta \det[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_k^{(2)}, \boldsymbol{a}_{k+1}, \cdots, \boldsymbol{a}_n]\end{aligned}$$

2. If for some $k$ we have $\boldsymbol{a}_k = \boldsymbol{a}_{k+1}$, then

$$\det A = \det[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_k, \boldsymbol{a}_{k+1}, \cdots, \boldsymbol{a}_n] = 0$$

3. Let

$$I_n = [\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n] = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

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$$\det I_n = 1$$

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Then

$$\det I_n = 1$$

**其它性质**:

$$\begin{aligned}
&\det[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_{k-1}, \boldsymbol{a}_k + \alpha \boldsymbol{a}_j, \boldsymbol{a}_{k+1}, \cdots, \boldsymbol{a}_j, \cdots, \boldsymbol{a}_n] \\
&= \det[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_k, \boldsymbol{a}_{k+1}, \cdots, \boldsymbol{a}_n] + \alpha \det[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_j, \cdots, \boldsymbol{a}_j, \cdots, \boldsymbol{a}_n] \\
&= \det[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n]
\end{aligned}$$

# 2.2. Rank of a Matrix

A $p$th-order minor of an $m \times n$ matrix $A$, with $p < \min\{m, n\}$, is the determinant of a $p \times p$ matrix obtained from $A$ by deleting $m - p$ rows and $n - p$ columns.

# 2.2. Rank of a Matrix

**Proposition 2.4** If an $m \times n$ $(m > n)$ matrix $A$ has a nonzero $n$-th-order minor, then the columns of $A$ are linearly independent; that is, $\operatorname{rank} A = n$.

**Proof**. Suppose that $A$ has a nonzero $n$th-order minor. Without loss of generality, we assume that the $n$th-order minor corresponding to the first $n$ rows of $A$ is nonzero. Let $x_i$, $i = 1, ..., n$, be scalars such that

$$x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n = 0.$$

i.e.,

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\
&\vdots \\
a_{n1}x_1 + a_{22}x_2 + \cdots + a_{nn}x_n &= 0 \\
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记 $\tilde{\boldsymbol{a}}_i = \begin{bmatrix} a_{1i} \\ \vdots \\ a_{ni} \end{bmatrix}$, then $x_1\tilde{\boldsymbol{a}}_1 + \cdots + x_n\tilde{\boldsymbol{a}}_n = \mathbf{0}$.

The $n$th-order minor is $\det[\tilde{\boldsymbol{a}}_1, , ..., \tilde{\boldsymbol{a}}_n]$, assumed to be nonzero. From the properties of determinants it follows that the columns $\tilde{\boldsymbol{a}}_1, \cdots, \tilde{\boldsymbol{a}}_n$ are linearly independent. Therefore, all $x_i = 0$, $i = 1, ..., n$. Hence, the columns $\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n$ an are linearly independent. □

# 2.2. Rank of a Matrix

**非异阵**

**对方阵** $A_{n\times n}$, $A$ **非异** iff $\exists B_{n\times n}$, such that

$$AB = BA = I_n.$$

$B$ **称为** $A$ **的逆阵, 记为** $B = A^{-1}$.

# 2.3. Linear Equations

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
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\end{aligned}
$$

i.e.,

$$x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n = \boldsymbol{b}.$$

i.e.,

$$A\boldsymbol{x} = \boldsymbol{b}$$

增广矩阵 $[A, \boldsymbol{b}] = [\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n, \boldsymbol{b}]$

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# 2.3. Linear Equations

**Theorem 2.1** The system of equations $A\boldsymbol{x} = \boldsymbol{b}$ has a solution if and only if

$$\operatorname{rank} A = \operatorname{rank}[A, \boldsymbol{b}].$$

**Proof**. $\Rightarrow$ Suppose that the system $A\boldsymbol{x} = \boldsymbol{b}$ has a solution. Therefore, $\boldsymbol{b}$ is a linear combination of the columns of $A$; that is, there exist $x_1, ..., x_n$ such that $x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n = \boldsymbol{b}$, 即 $\boldsymbol{b} \in \operatorname{span}[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n]$. 因此

$$\begin{aligned} \operatorname{rank} A &= \dim \operatorname{span}[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n] \\ &= \dim \operatorname{span}[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n, \boldsymbol{b}] \\ &= \operatorname{rank}[A, \boldsymbol{b}] \end{aligned}$$

# 2.3. Linear Equations

**Theorem 2.1** The system of equations $A\boldsymbol{x} = \boldsymbol{b}$ has a solution if and only if

$$\operatorname{rank} A = \operatorname{rank}[A, \boldsymbol{b}].$$

**Proof**. $\Rightarrow$ Suppose that the system $A\boldsymbol{x} = \boldsymbol{b}$ has a solution. Therefore, $\boldsymbol{b}$ is a linear combination of the columns of $A$; that is, there exist $x_1, ..., x_n$ such that $x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n = \boldsymbol{b}$, 即 $\boldsymbol{b} \in \operatorname{span}[\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n]$. 因此

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$\Leftarrow$ Suppose that $\operatorname{rank} A = \operatorname{rank}[A, \boldsymbol{b}] = r$. Thus, we have $r$ linearly independent columns of $A$. Without loss of generality, let $\boldsymbol{a}_1, ..., \boldsymbol{a}_r$ be these columns. Therefore, $\boldsymbol{a}_1, ..., \boldsymbol{a}_r$ are also linearly independent columns of the matrix $[A, \boldsymbol{b}]$. Because $\operatorname{rank}[A, \boldsymbol{b}] = r$, the remaining columns of $[A, \boldsymbol{b}]$ can be expressed as linear combinations of $\boldsymbol{a}_1, ..., \boldsymbol{a}_r$. In particular, $\boldsymbol{b}$ can be expressed as a linear combination of these columns. Hence, there exist $x_1, ..., x_n$ such that $x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n = \boldsymbol{b}$. □

# 2.3. Linear Equations

**Theorem 2.2** Consider the equation $A\boldsymbol{x} = \boldsymbol{b}$, where $A \in \mathbb{R}^{m\times n}$ and $\operatorname{rank} A = m$. A solution to $A\boldsymbol{x} = \boldsymbol{b}$ can be obtained by assigning arbitrary values for $n - m$ variables and solving for the remaining ones.

**Proof**. We have $\operatorname{rank} A = m$, and therefore we can find $m$ linearly independent columns of $A$. Without loss of generality, let $\boldsymbol{a}_1, ..., \boldsymbol{a}_m$ be such columns. Rewrite the equation $A\boldsymbol{x} = \boldsymbol{b}$ as

$$x_1\boldsymbol{a}_1 + x_m\boldsymbol{a}_m = \boldsymbol{b} - x_{m+1}\boldsymbol{a}_{m+1} - \cdots - x_n\boldsymbol{a}_n.$$

Assign to $x_{m+1}$, $x_{m+2}$, ..., $x_n$ arbitrary values, say

$$x_{m+1} = d_{m+1}, x_{m+2} = d_{m+2}, \cdots, x_n = d_m,$$

and let

$$B = [\boldsymbol{a}_1, \cdots, \boldsymbol{a}_m] \in \mathbb{R}^{m\times m}.$$

## 2.3. Linear Equations

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# 2.3. Linear Equations

Note that $\det B \neq 0$. We can represent the system of equations as

$$B \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} = [\boldsymbol{b} - d_{m+1}\boldsymbol{a}_{m+1} - \cdots - d_n\boldsymbol{a}_n].$$

因为 $B$ 可逆, 所以

$$\begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = B^{-1}[\boldsymbol{b} - d_{m+1}\boldsymbol{a}_{m+1} - \cdots - d_n\boldsymbol{a}_n].$$

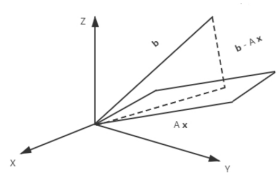
□

# 思考题

$$\begin{cases} x+y=2 \\ x-y=0 \\ x+2y=4 \end{cases}$$

# 线性方程组的求解：投影的例子

$$\begin{cases} x+y=2 \\ x-y=0 \\ x+2y=4 \end{cases} \quad \text{归结为} \quad \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix}$$




$$A\boldsymbol{x} = \boldsymbol{b}$$
$$A^\top A\boldsymbol{x} = A^\top \boldsymbol{b}$$
$$\boldsymbol{x} = \left(A^\top A\right)^{-1} A^\top \boldsymbol{b}$$

# 线性方程组的求解：投影的例子

给了一个向量 $\boldsymbol{b}$，和一个平面（由 $\boldsymbol{a}_1$ 和 $\boldsymbol{a}_2$ 生成），要求 $P$，使 $(P\boldsymbol{b})$ 在平面内，i.e., $\boldsymbol{b} = P\boldsymbol{b} + (\boldsymbol{b} - p\boldsymbol{b})$，且 $(P\boldsymbol{b})^\top \cdot (\boldsymbol{b} - P\boldsymbol{b}) = 0$.

如何求：从上张 Slide，$\boldsymbol{b}$ 就是上张的 $\boldsymbol{b}$，$A = (\boldsymbol{a}_1, \boldsymbol{a}_2)$，且：$A\boldsymbol{x} = \boldsymbol{b}$ 的最优解（误差最小解）$x^* = \left(A^\top A\right)^{-1} A^\top \boldsymbol{b}$, 而 $A\boldsymbol{x}^*$ 就是 $P\boldsymbol{b}$. $\therefore A\boldsymbol{x}^* = A\left(A^\top A\right)^{-1} A^\top \boldsymbol{b}$. 由 $A\boldsymbol{x}^* = P\boldsymbol{b}$ 得 $P = A\left(A^\top A\right)^{-1} A^\top$

下面验证：

$$(P\boldsymbol{b})^\top (\boldsymbol{b} - P\boldsymbol{b}) = 0$$

验证：

$$\begin{aligned}
(P\boldsymbol{b})^\top (\boldsymbol{b} - P\boldsymbol{b}) &= (A\boldsymbol{x}^*)^\top (\boldsymbol{b} - A\boldsymbol{x}^*) \\
&= \boldsymbol{x}^{*\top} A^\top \boldsymbol{b} - \boldsymbol{x}^{*\top} A^\top A\boldsymbol{x}^* \\
&\overset{x^* = \left(A^\top A\right)^{-1} A^\top b}{=\!=} \boldsymbol{x}^{*\top} A^\top \boldsymbol{b} - \boldsymbol{x}^{*\top} A^\top A\left(A^\top A\right)^{-1} A^\top \boldsymbol{b} \\
&= \left(\boldsymbol{x}^{*\top} A^\top\right) \boldsymbol{b} - \left(\boldsymbol{x}^{*\top} A^\top\right) \boldsymbol{b} = 0 \\
&= \left(\boldsymbol{x}^{*\top} \mathrm{A}^\top - \boldsymbol{x}^{*\top} \mathrm{A}^\top\right) \boldsymbol{b} = 0
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## 2.4 Inner Products and Norms

**对于绝对值, 有以下的公式:**

1. $|a| = |-a|$.
2. $-|a| \leq a < |a|$.
3. $|a+b| < |a| + |b|$.
4. $||a| - |b|| < |a-b| < |a| + |b|$.
5. $|ab| = |a|\,|b|$.
6. $|a| < c$ and $|b| < d$ imply that $|a+b| < c+d$.
7. The inequality $|a| < b$ is equivalent to $-b < a < b$ (i.e., $a < b$ and $-a < b$). The same holds if we replace every occurrence of "$<$" by "$\leq$"

# 2.4 Inner Products and Norms

For $\boldsymbol{x}$, $\boldsymbol{y} \in \mathbb{R}^n$, we define the Euclidean inner product by

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \sum_{i=1}^{n} x_i y_i = \boldsymbol{x}^\top \boldsymbol{y}.$$

**内积具有以下性质**:

1. Positivity: $\langle \boldsymbol{x}, \boldsymbol{x} \rangle > 0$, $\langle \boldsymbol{x}, \boldsymbol{x} \rangle = 0$ iff $\boldsymbol{x} = 0$.
2. Symmetry: $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle$.
3. Additivity: $\langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle = \langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle$.
4. Homogeneity (**齐性**): $\langle r\boldsymbol{x}, \boldsymbol{y} \rangle = r\langle \boldsymbol{x}, \boldsymbol{y} \rangle$ for every $r \in \mathbb{R}$.

The properties of additivity and homogeneity in the second vector also hold;

$$\langle \boldsymbol{x}, \boldsymbol{y} + \boldsymbol{z} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle + \langle \boldsymbol{x}, \boldsymbol{z} \rangle,$$
$$\langle \boldsymbol{x}, r\boldsymbol{y} \rangle = r\langle \boldsymbol{x}, \boldsymbol{y} \rangle \text{ for every } r \in \mathbb{R}.$$

# 2.4 Inner Products and Norms

The vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ are said to be orthogonal if $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$.

The Euclidean norm of a vector $\boldsymbol{x}$ is defined as

$$\|\boldsymbol{x}\| = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle} = \sqrt{\boldsymbol{x}^\top \boldsymbol{x}}$$

**Theorem 2.3 Cauchy-Schwarz Inequality**. For any two vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ in $\mathbb{R}^n$, the Cauchy-Schwarz inequality

$$|\langle \boldsymbol{x}, \boldsymbol{y} \rangle| \leq \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|$$

holds. Furthermore, equality holds iff $\boldsymbol{x} = \alpha \boldsymbol{y}$ for some $\alpha \in \mathbb{R}$.

**Proof**. First assume that $\boldsymbol{x}$ and $\boldsymbol{y}$ are unit vectors; that is, $\|\boldsymbol{x}\| = \|\boldsymbol{y}\| = 1$. Then,

$$\begin{aligned} 0 \leq \|\boldsymbol{x} - \boldsymbol{y}\|^2 &= \langle \boldsymbol{x} - \boldsymbol{y}, \boldsymbol{x} - \boldsymbol{y} \rangle \\ &= \|\boldsymbol{x}\|^2 - 2\langle \boldsymbol{x}, \boldsymbol{y} \rangle + \|\boldsymbol{y}\|^2 \\ &= 2 - 2\langle \boldsymbol{x}, \boldsymbol{y} \rangle \end{aligned}$$

or

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle \leq 1,$$

等号成立当且仅当 $\boldsymbol{x} = \boldsymbol{y}$.

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# 2.4 Inner Products and Norms

Next, assuming that neither $\boldsymbol{x}$ nor $\boldsymbol{y}$ is zero (for the inequality obviously holds if one of them is zero), we replace $\boldsymbol{x}$ and $\boldsymbol{y}$ by the unit vectors $\boldsymbol{x}/\|\boldsymbol{x}\|$ and $\boldsymbol{y}/\|\boldsymbol{y}\|$. Then

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle \leq \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|.$$

Replacing $\boldsymbol{x}$ by $-\boldsymbol{x}$, we have

$$-\langle \boldsymbol{x}, \boldsymbol{y} \rangle \leq \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|.$$

**所以**,

$$|\langle \boldsymbol{x}, \boldsymbol{y} \rangle| \leq \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|$$

**等号成立当且仅当** $\boldsymbol{x}/\|\boldsymbol{x}\| = \pm \boldsymbol{y}/\|\boldsymbol{y}\|$; that is, $\boldsymbol{x} = \alpha \boldsymbol{y}$ for some $\alpha \in \mathbb{R}$.

# 2.4 Inner Products and Norms

The Euclidean norm of a vector $\|\boldsymbol{x}\|$ has the following properties:

1. Positivity: $\|\boldsymbol{x}\| > 0$, $\|\boldsymbol{x}\| = \mathbf{0}$ iff $\boldsymbol{x} = \mathbf{0}$.

2. Homogeneity: $\|r\boldsymbol{x}\| = |r| \cdot \|\boldsymbol{x}\|$, $r \in \mathbb{R}$.

3. Triangle inequality: $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$.

三角不等式可用 Cauchy-Schwarz 不等式证明:

$$\begin{aligned} \|\boldsymbol{x} + \boldsymbol{y}\|^2 &\overset{def}{=\!=} \|\boldsymbol{x}\|^2 + 2\langle \boldsymbol{x}, \boldsymbol{y} \rangle + \|\boldsymbol{y}\|^2 \\ &\overset{C-S}{\leq} \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\|\|\boldsymbol{y}\| + \|\boldsymbol{y}\|^2 \\ &= (\|\boldsymbol{x}\| + \|\boldsymbol{y}\|)^2 \end{aligned}$$

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# 2.4 Inner Products and Norms

$p$-norm

**刚才的 Euclidean norm 是以下形式的 norm 取 $p = 2$ 的情形**

$$\|\boldsymbol{x}\|_p = \begin{cases} (|x_1|^p + \cdots + |x_n|^p)^{1/p} & \text{if } 1 \le p < \infty \\ \max\{|x_1|, ..., |x_n|\} & \text{if } p = \infty \end{cases}$$

---

Norm 可以用来描述连续函数.

A function $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ is continuous at $\boldsymbol{x}$ if for all $\varepsilon > 0$, there exists $\delta > 0$ such that $\|\boldsymbol{y} - \boldsymbol{x}\| < \delta \Rightarrow \|\boldsymbol{f}(\boldsymbol{y}) - \boldsymbol{f}(\boldsymbol{x})\| < \varepsilon$.

If the function $\boldsymbol{f}$ is continuous at every point in $\mathbb{R}^n$, we say that it is continuous on $\mathbb{R}^n$. Note that $\boldsymbol{f} = [f_1, ..., f_m]^\top$ is continuous iff each component $f_i$, $i = 1, ..., m$, is continuous

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## 2.4 Inner Products and Norms

For the complex vector space $\mathbb{C}^n$, we define an inner product $\langle \boldsymbol{x}, \boldsymbol{y} \rangle$ to be $\sum_{i=1}^{n} x_i \bar{y}_i$.

The inner product on $\mathbb{C}^n$ is a complex-valued function having the following properties:

1. Positivity: $\langle \boldsymbol{x}, \boldsymbol{x} \rangle > 0$, $\langle \boldsymbol{x}, \boldsymbol{x} \rangle = 0$ iff $\boldsymbol{x} = 0$.

2. Symmetry: $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle}$.

3. Additivity: $\langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle = \langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle$.

4. Homogeneity (**斉性**): $\langle r\boldsymbol{x}, \boldsymbol{y} \rangle = r\langle \boldsymbol{x}, \boldsymbol{y} \rangle$ for every $r \in \mathbb{C}$.

Also,

$$\langle \boldsymbol{x}, r_1\boldsymbol{y} + r_2\boldsymbol{z} \rangle = \bar{r}_1 \langle \boldsymbol{x}, \boldsymbol{y} \rangle + \bar{r}_2 \langle \boldsymbol{x}, \boldsymbol{z} \rangle.$$

# 回家作业（不需提交）

给出 $\begin{cases} x+y=2 \\ x-y=0 \\ x+2y=4 \end{cases}$ 的最优解的两种方法的等价证明。